Module 01 Case Study: Enterprise Data Governance Foundation

Case Title: “RetailCorp: The Data Chaos Challenge”

1. Background Context

RetailCorp is a **mid-to-large global retail organization** operating across:

- E-commerce platforms
- Physical stores
- Mobile applications
- Third-party marketplaces

Over the past 5 years, the company has rapidly expanded its digital footprint, resulting in **massive data growth** across multiple systems.

🔶 Current Systems Landscape

System	Description	Data Stored
CRM System	Customer relationship management	Customer profiles
POS System	In-store transactions	Sales transactions
E-commerce Platform	Online orders	Orders, clickstream
Marketing System	Campaign tracking	Leads, engagement
Data Lake (new)	Raw data storage	Mixed structured/unstructured

🔶 Key Problem Statement

RetailCorp leadership has identified **critical data issues**:

❌ *Problem 1: Inconsistent Definitions*

- "Customer" is defined differently across systems:
  - CRM → registered user
  - POS → buyer
  - Marketing → email subscriber

❌ *Problem 2: Lack of Ownership*

- No clear:
  - Data owners
  - Data stewards
- Teams modify data independently

❌ *Problem 3: Poor Data Trust*

- Reports from different teams show **different numbers**
- Leadership has **lost confidence in analytics**

❌ *Problem 4: No Metadata Visibility*

- No central catalog
- Analysts don't know:
  - Where data lives
  - What it means
  - Who owns it

❌ *Problem 5: Compliance Risk*

- Personally identifiable information (PII) exists without:
    - Classification
    - Governance policies

2. Business Impact

These issues have led to:

- 📉 Incorrect business decisions
- 💰 Revenue leakage (duplicate targeting, missed customers)
- ⚠️ Regulatory risk (GDPR/CCPA exposure)
- ⏱️ Increased time to find and trust data

3. Objective of This Case Study

You are hired as a **Data Governance Consulting Team**.

Your mission:

Design and implement a **foundational data governance framework** for RetailCorp.

4. Tasks to Be Accomplished

Task 1: Define Data Governance Structure

Requirements:

- Identify:
    - Data Owners
    - Data Stewards
    - Data Consumers

Output:

- Governance organizational chart

- Roles & responsibilities table

Task 2: Create Business Glossary

Requirements:

Define standardized terms:

- Customer
- Order
- Revenue
- Product

Use:

- OpenMetadata

Output:

- Business glossary entries
- Term definitions with ownership

Task 3: Data Asset Registration

Requirements:

- Register datasets (sample CSVs provided)
- Define:
    - Table names
    - Column descriptions

Use:

- OpenMetadata

Output:

- Metadata catalog entries

Task 4: Identify Data Domains

Requirements:

Group data into domains:

- Customer Data
- Sales Data
- Marketing Data

Output:

- Domain classification diagram

Task 5: Define Governance Policies (Conceptual)

Requirements:

Create policies for:

- Data ownership
- Data access
- Data definition standardization

Output:

- Governance policy document

Task 6: Initial Data Exploration (Optional Technical)

Requirements:

- Load sample dataset into:
    - Databricks Free Edition
- Explore:
    - Columns
    - Missing values

Output:

- Basic data profiling summary

5. Data Provided to you

Dataset 1: Customers.csv

- customer_id
- name
- email
- signup_date

Dataset 2: Orders.csv

- order_id
- customer_id
- order_amount
- order_date

Dataset 3: Marketing.csv

- email
- campaign_id
- engagement_score

6. Expected Outcomes

By completing this case study, you will:

- Understand **why data governance is needed**
- Learn **how to define business terms**
- Identify **organizational roles in governance**
- Gain hands-on exposure to:
    - Metadata management
    - Data cataloging
- Recognize **real-world enterprise challenges**

Module 01 Lab Manual

Lab Title: Building a Data Catalog and Business Glossary for RetailCorp

1. Lab Objective

In this lab, you will:

- Create a **data catalog**
- Define a **business glossary**
- Assign **data ownership**
- Register datasets and metadata
- Perform **basic data exploration**

2. Prerequisites

Before starting:

Accounts Required

- Databricks Free Edition account (instructions were given in coming sections)
- Local setup or hosted access to Open Metadata (instructions were given in coming sections)

Software Required

- Web browser
- CSV datasets (provided by instructor)

3. Dataset Description

You will use:

Customers.csv

- customer_id
- name
- email
- signup_date

Orders.csv

- order_id
- customer_id
- order_amount
- order_date

4. Lab Tasks Overview

Task	Tool
Upload & explore data	Databricks
Register datasets	OpenMetadata
Create glossary	OpenMetadata
Assign ownership	OpenMetadata

5. Part A: Data Exploration in Databricks

Step 1: Login

- Go to Databricks Free Edition https://login.databricks.com/
- Log in with your account

Step 2: Create a Notebook

- Click **Workspace → Create → Notebook**

- Name: Module1_Data_Exploration
- Language: Python or SQL

Step 3: Upload Dataset

- Go to **Data → Add Data**
- Upload:
  - Customers.csv
  - Orders.csv

Step 4: Load Data

Python Example:

```
customers = spark.read.csv("/FileStore/tables/Customers.csv", header=True,
inferSchema=True)
orders = spark.read.csv("/FileStore/tables/Orders.csv", header=True, inferSchema=True)
```

Step 5: Explore Data

```
customers.show()
customers.printSchema()

orders.show()
orders.printSchema()
```

Step 6: Basic Profiling

```
customers.describe().show()
orders.describe().show()
```

Observe:

- Missing values
- Data types

- Data distribution

Output

- Screenshot of schema
- Summary observations

6. Part B: Setup in OpenMetadata

Install it using the instructions https://docs.open-metadata.org/v1.12.x/deployment

Step 1: Launch Tool

- Open OpenMetadata in browser http://localhost:8585

Or

Use Docker:

git clone https://github.com/open-metadata/OpenMetadata

cd OpenMetadata

docker compose up -d

Step 2: Create a Service

- Go to **Settings → Services → Add Service**
- Choose:
    - Database Service (e.g., MySQL / sample DB)

If DB connection is not available:

- Use **sample/demo service**

Step 3: Add Database & Tables

- Navigate to:
    - Database → Schema → Tables

- Go to:
    - **Explore → Tables**
- Click **Add Table**
- Create:

- Create entries manually if needed:
    - Customers
    - Orders

7. Part C: Create Business Glossary

Step 1: Navigate to Glossary

- Click **Govern → Glossary**

Step 2: Create Glossary

- Name: RetailCorp Glossary

Step 3: Add Terms

Create the following:

Term 1: Customer

- Definition: “An individual who has purchased a product from RetailCorp”

Term 2: Order

- Definition: “A transaction representing a purchase made by a customer”

Term 3: Revenue

- Definition: “Total monetary value generated from orders”

Step 4: Link Terms to Tables

- Map:

    - Customer → Customers table
    - Order → Orders table

Output

- Screenshot of glossary
- Defined terms

8. Part D: Assign Data Ownership

Step 1: Create Users (if needed)

- Go to **Settings → Users**

Step 2: Assign Roles

Example:

- Data Owner → Sales Manager
- Data Steward → Data Analyst

Step 3: Assign Ownership

- Go to table:
    - Customers
- Assign:
    - Owner
    - Steward

Output

- Ownership assignment screenshot

9. Part E: Metadata Enrichment

Step 1: Add Column Descriptions

Example:

Column	Description
customer_id	Unique identifier
email	Customer email

Step 2: Add Tags (Optional)

- Tag:
  - email → PII

Output

- Enriched metadata view

PART 3: Connecting Databricks (Conceptual Intro)

Explain (no need to fully implement yet):

OpenMetadata can ingest metadata from Databricks:

- Tables
- Lineage
- Queries

FINAL STUDENT DELIVERABLES

You must submit:

1. Databricks Notebook

- Data loaded

- Schema explored

2. Glossary

- At least 3 business terms

3. Metadata

- Tables + column descriptions

4. Ownership

- Defined roles

5. Reflection

- Why governance matters